

SUSTAINABILITY & CONNECTED AUTONOMY:

A New Era For Aviation

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TEAM



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- PhD Student Engineering and Public Policy
- Master's and Bachelor's in Operations Research



Iana Iacob

- PhD Student Engineering and Public Policy
- MBA Instituto de Empresa (Spain)
- Master's in Petroleum Engineering
- Bachelor's in Chemical Engineering



Ashima Sharma

- Bachelor's in Civil and Environmental Engineering
- Minors in HCI and Technology and Policy



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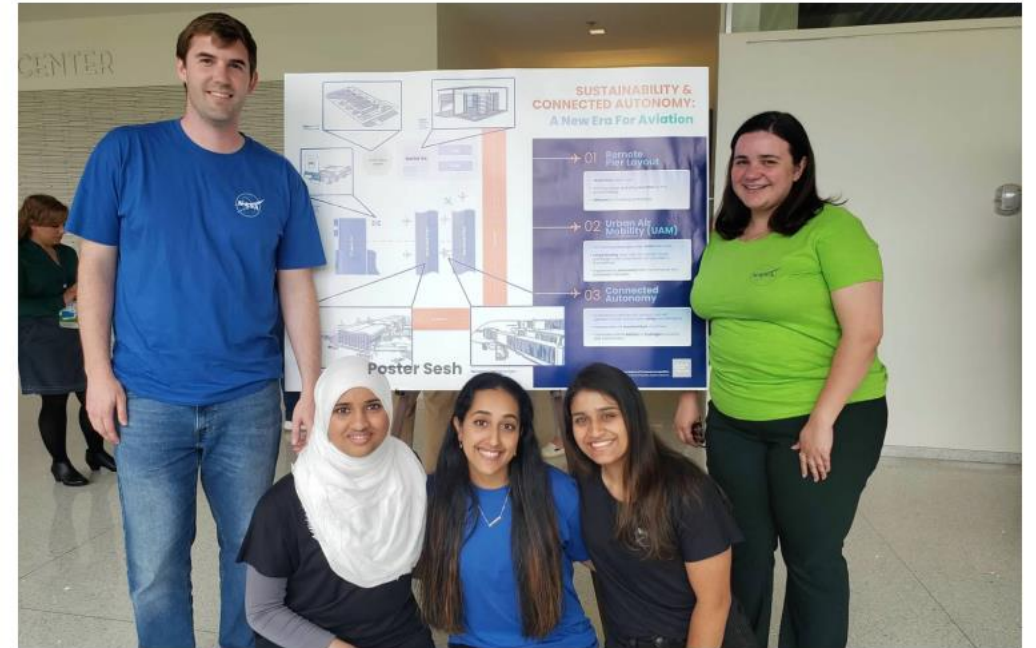
- Senior Communications Design major
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Tahaseen Shaik

- Junior Electrical & Computer Engineering, Robotics double major

NASA GATEWAYS TO BLUE SKIES COMPETITION



OVERVIEW



Airport Innovations

Gate Design
Remote Pier
Connected Autonomy
Landing Port
Centralized Energy Hub



Key Takeaways

Sustainability and Connected Autonomy



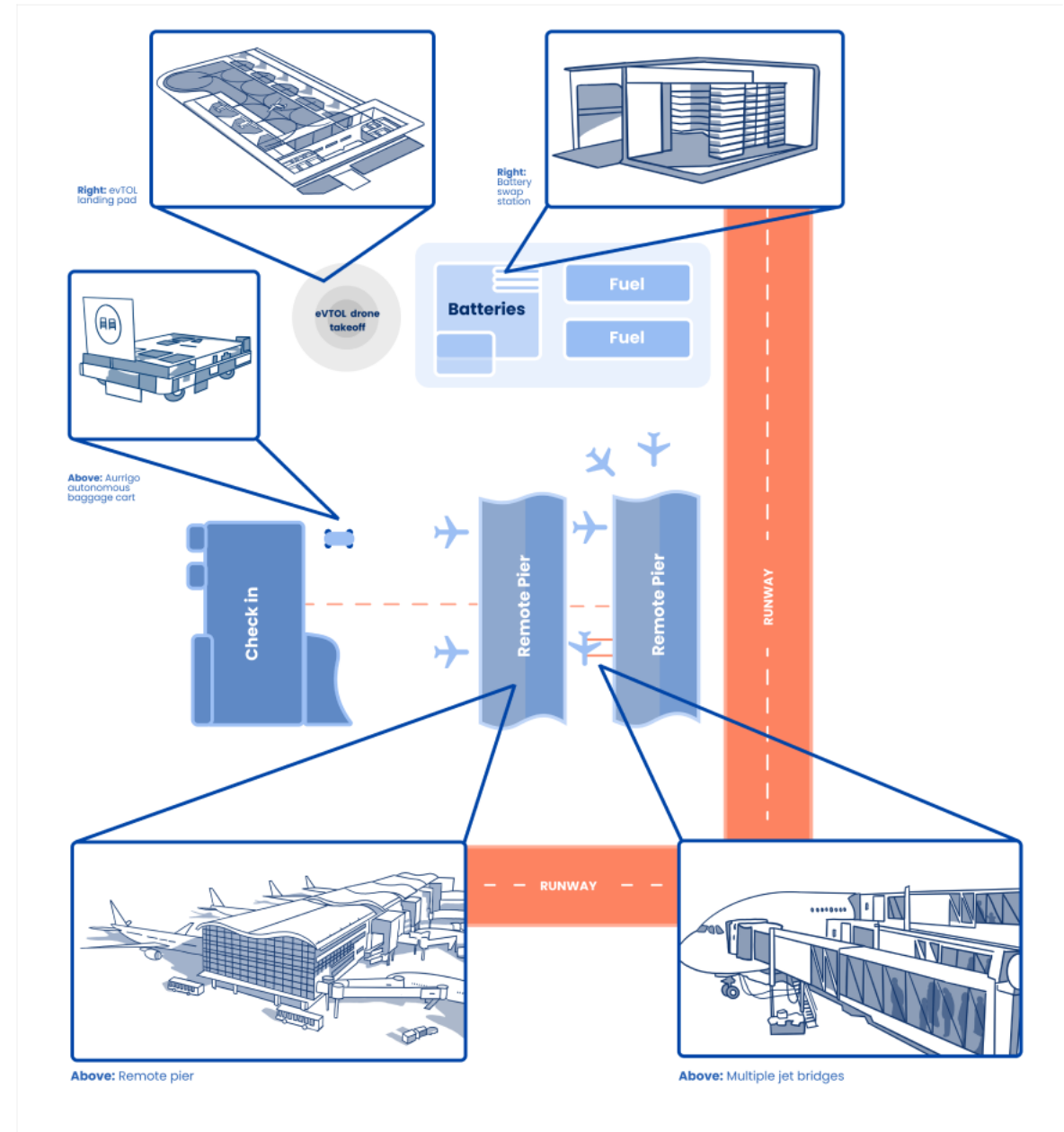
Denver Case Study



Technology Readiness



Conclusions



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Sustainability and Connected Autonomy



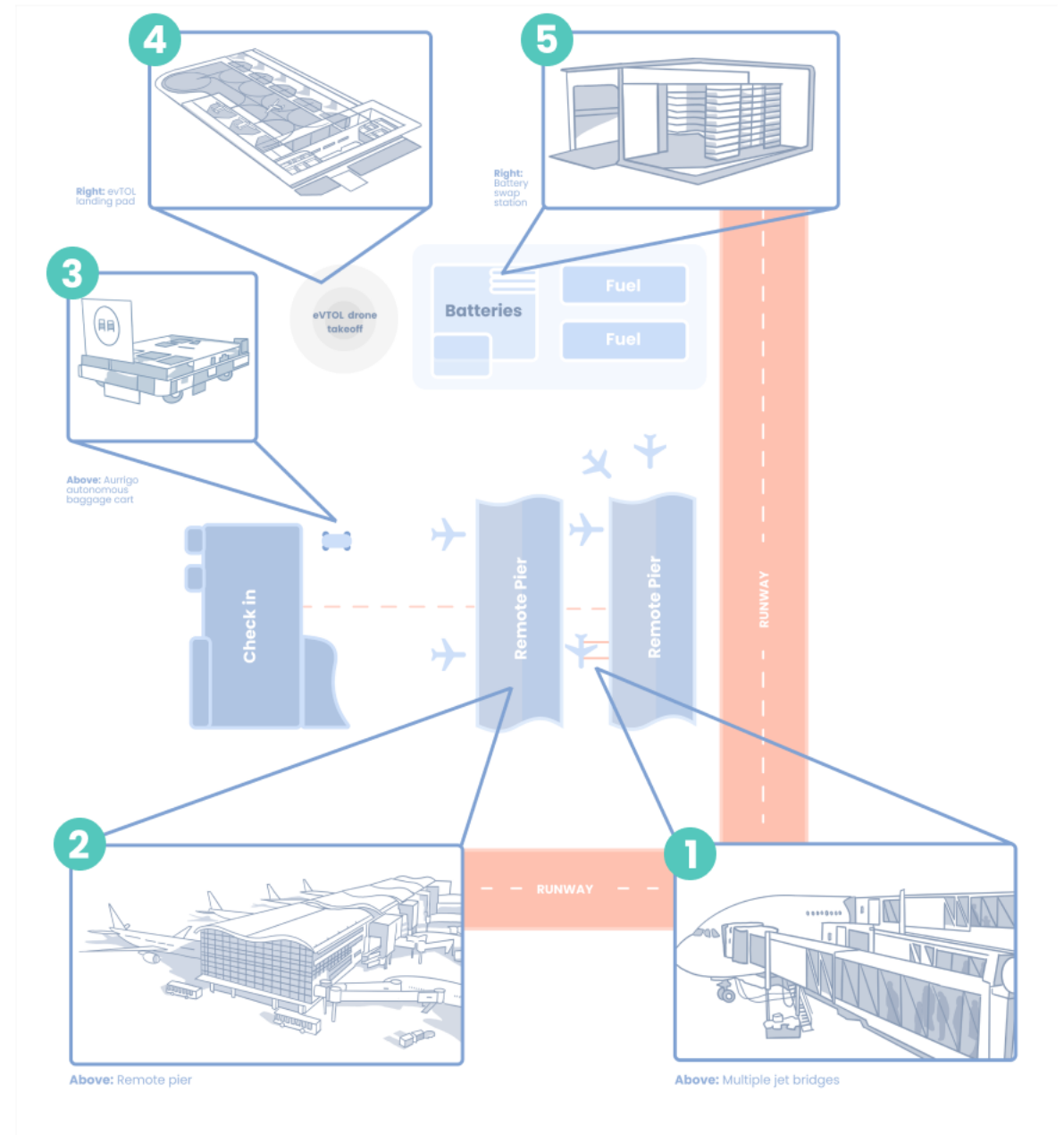
Denver Case Study



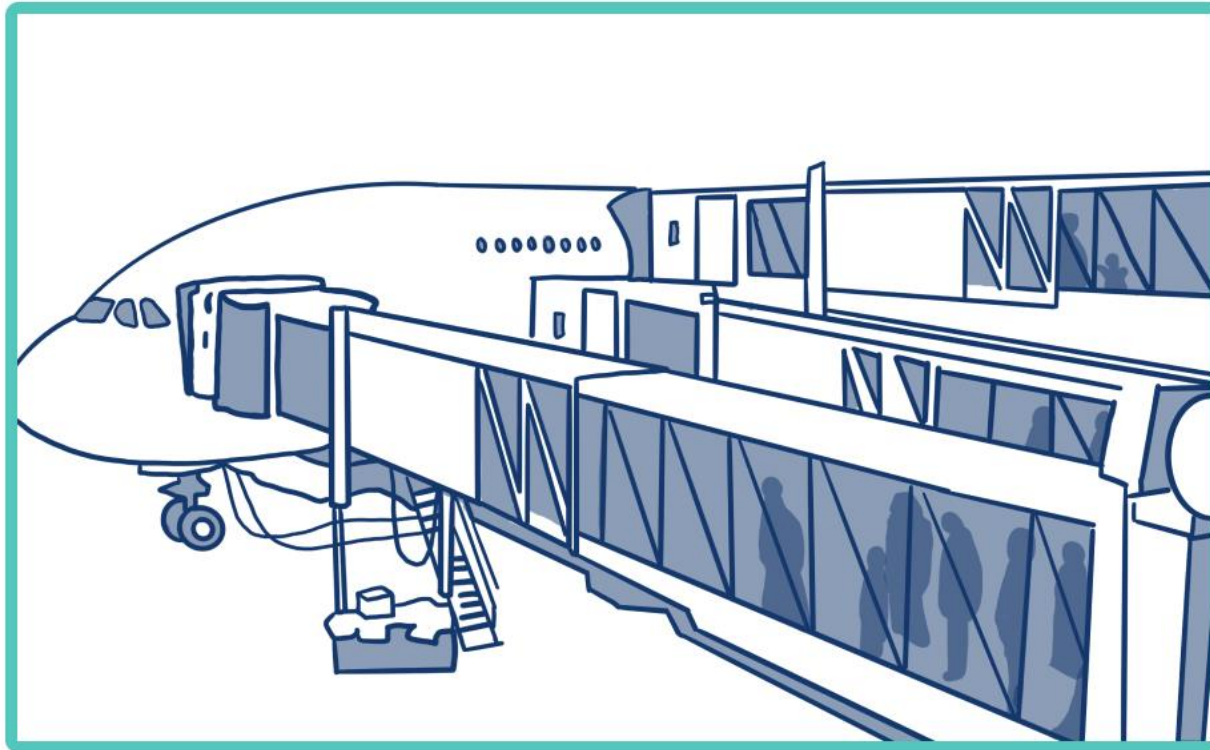
Technology Readiness



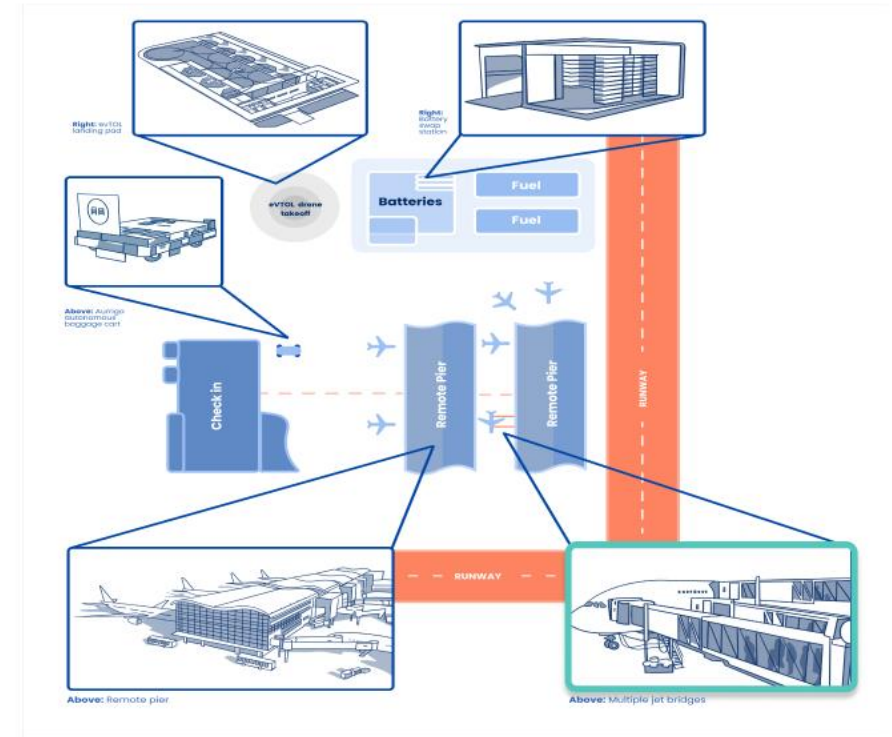
Conclusions



✈ Airport Innovations



Multiple jet bridges



✈ Aircraft Form

Blended Wing Body (BWB) Aircraft are versatile and allow for a great number of benefits:

- Blended wings **reduce** noise pollutions and maximize efficiency
- **Supports** both long-range flying and short-range/high-frequency flying
- **Quicker** certification process because already in use

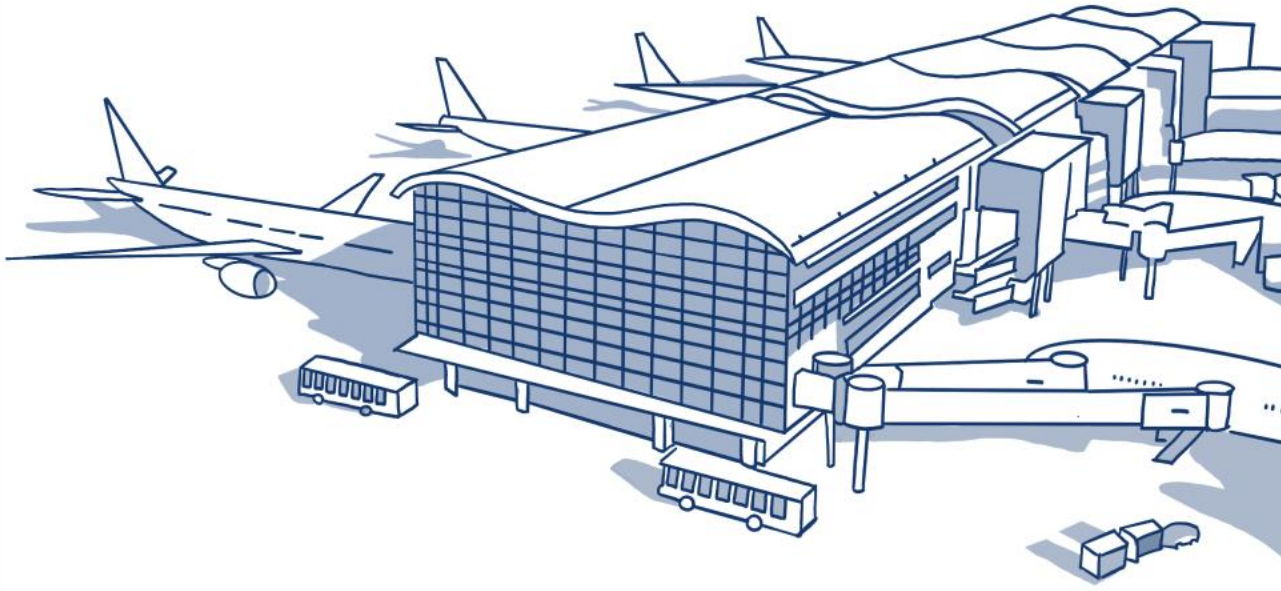
Supersonic aircraft will be used to support long range flights



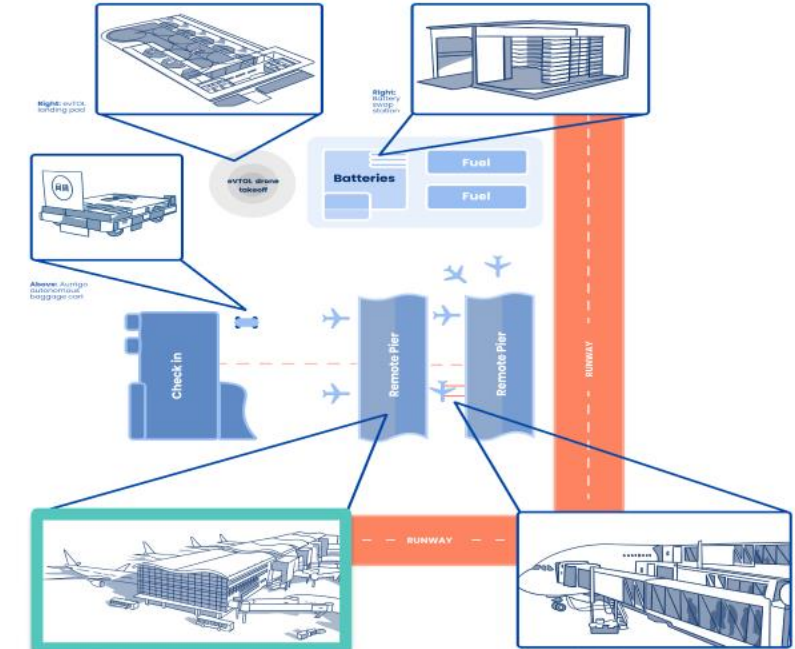
✈ Gate Design

- Blended wing-body aircraft can operate **out of** existing airport terminals and gates
- **Multiple** jet bridges can be useful for more efficient boarding
- Jet bridge configurations can be adapted to existing gates at airports, including DEN



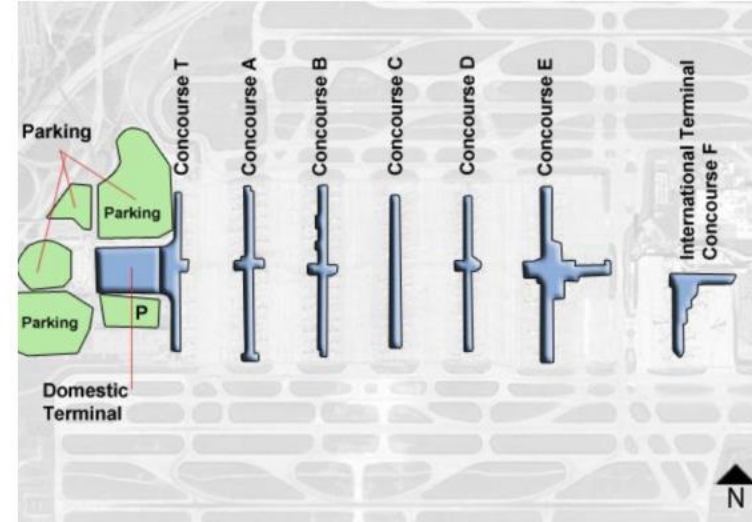


Remote Pier Layout

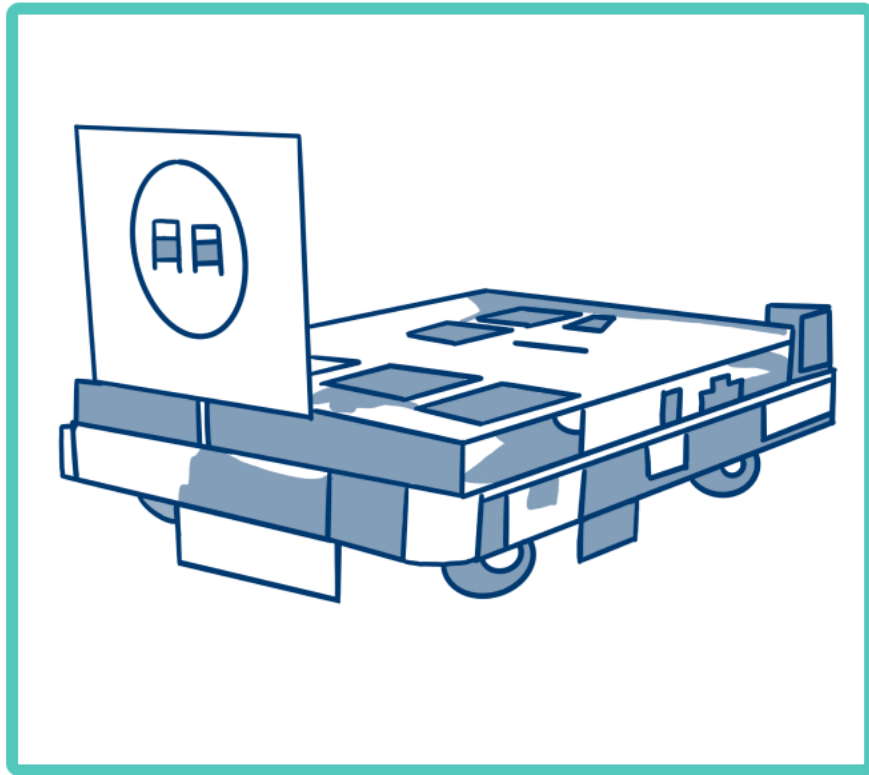


✈ Remote Pier

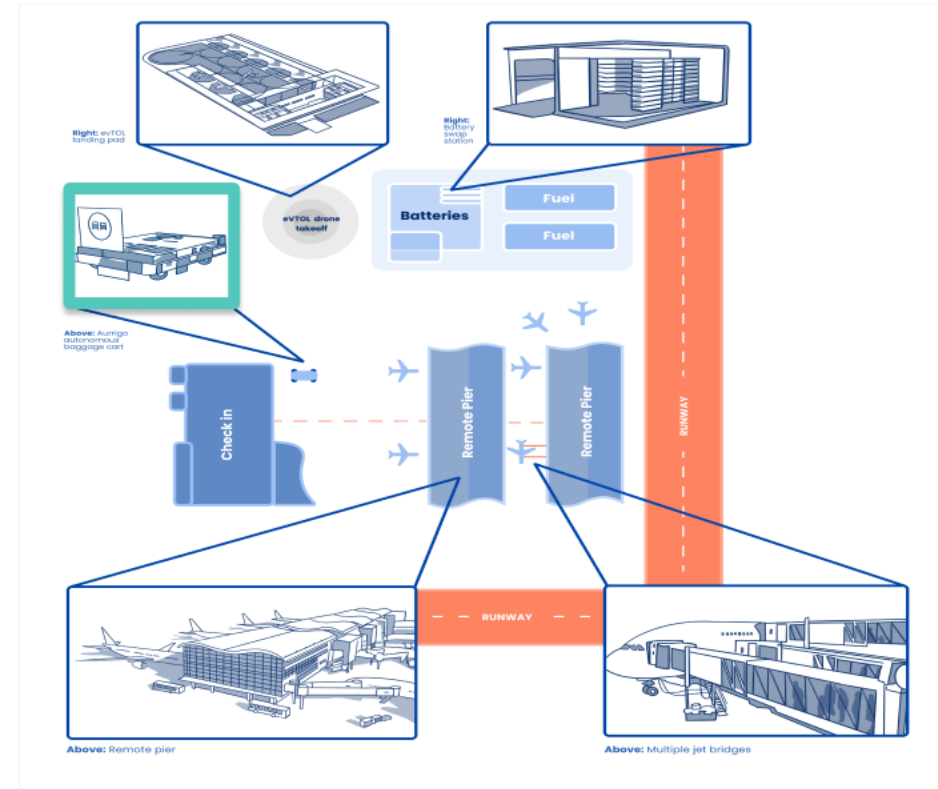
- **Maximizes** apron use
- Arriving planes spend **less time** on the ground
- **Efficient** for handling passengers
- Necessitates the use of a **sophisticated** people mover system



✈ Airport Innovations: Connected Autonomy



Aurrigo Autonomous baggage cart



✈ Integrated Autonomous Platform

A fully connected autonomous system to connect discrete services

- Reliance on **autonomous** vehicles to complete essential routine tasks:
 - Snow cleaning, bird mitigation, preflight inspection, etc.
- Allows airports to **recover** from natural disasters and turbulent weather quicker
- Primary risks include **cybersecurity** vulnerabilities

DEN is a prime location for a system like this because it's a prime logistics hub as well



Top Left: NASA, 2021

Top Right: Airport Technology, 2019

Bottom: Scandinavian Traveler, 2021

✈ Standardized Logistics Containers

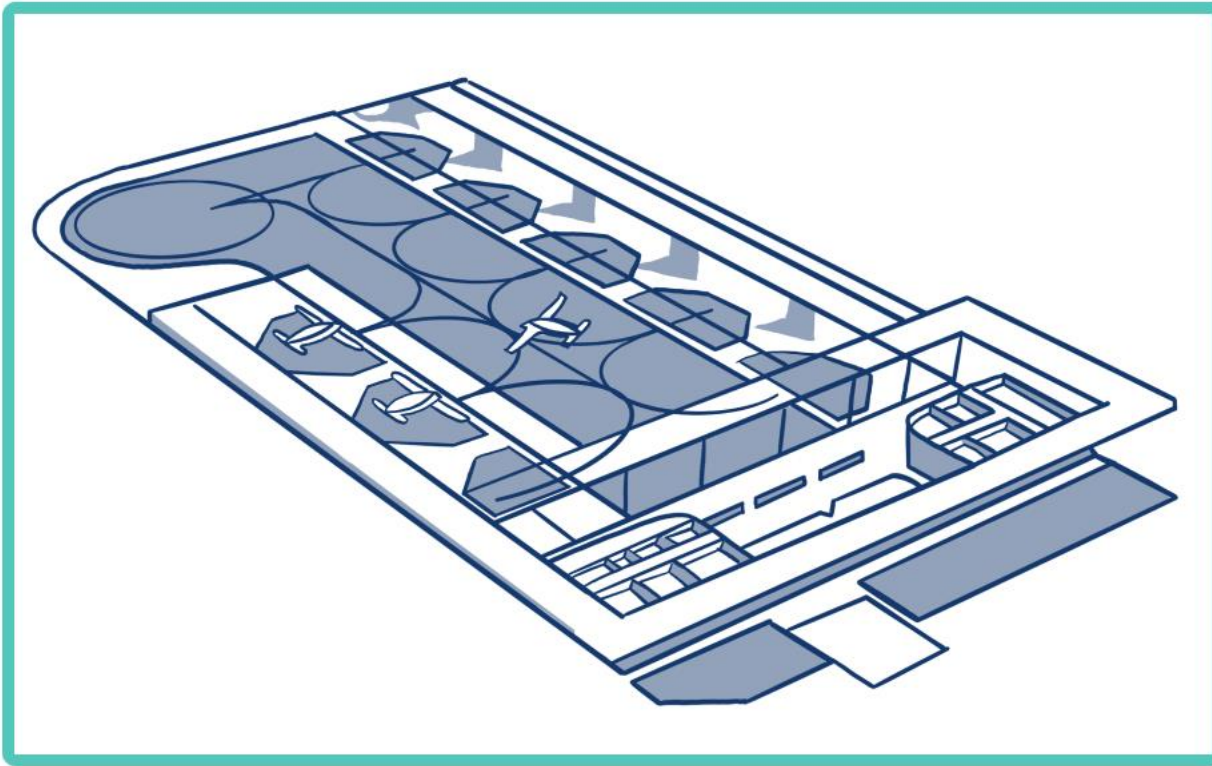
These logistics containers or AGV (automated guided vehicles) **allow for efficient transportation of goods** (baggage, fuel, etc.) throughout the airport

- Containers can be **automated** to follow set paths at set schedules
- Can be **restructured** to allow passenger travel
- Can be **employed in large quantities** to transport cargo quickly
- **Reduces** human burden and maximizes efficiency

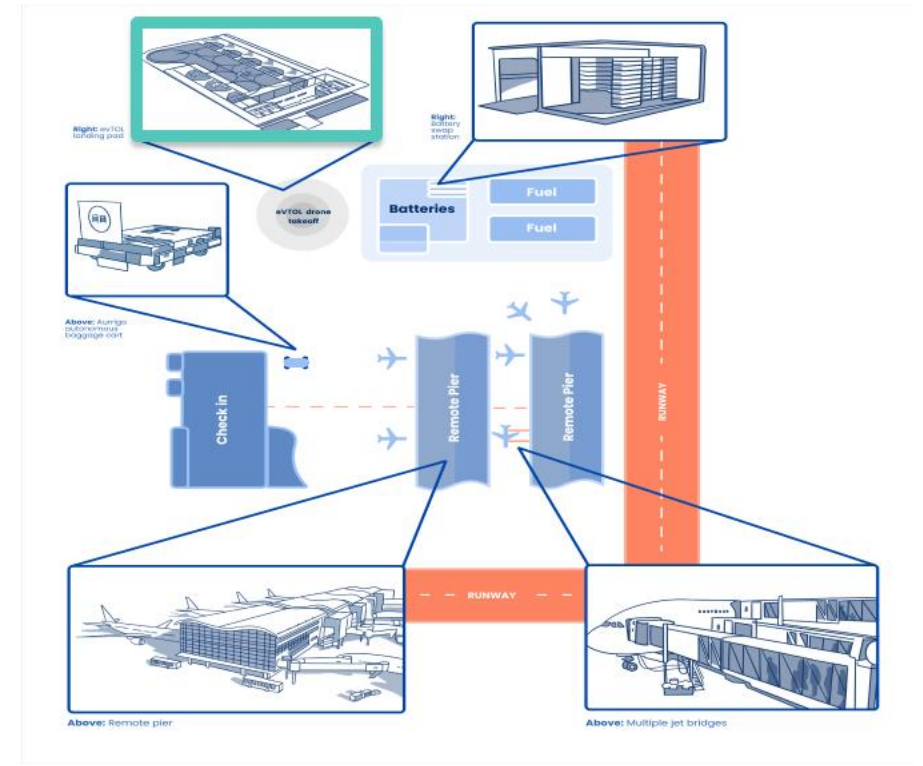


Top: Konecranes, 2022
Bottom: Aurigo, 2019

✈ Airport Innovations



Landing Port



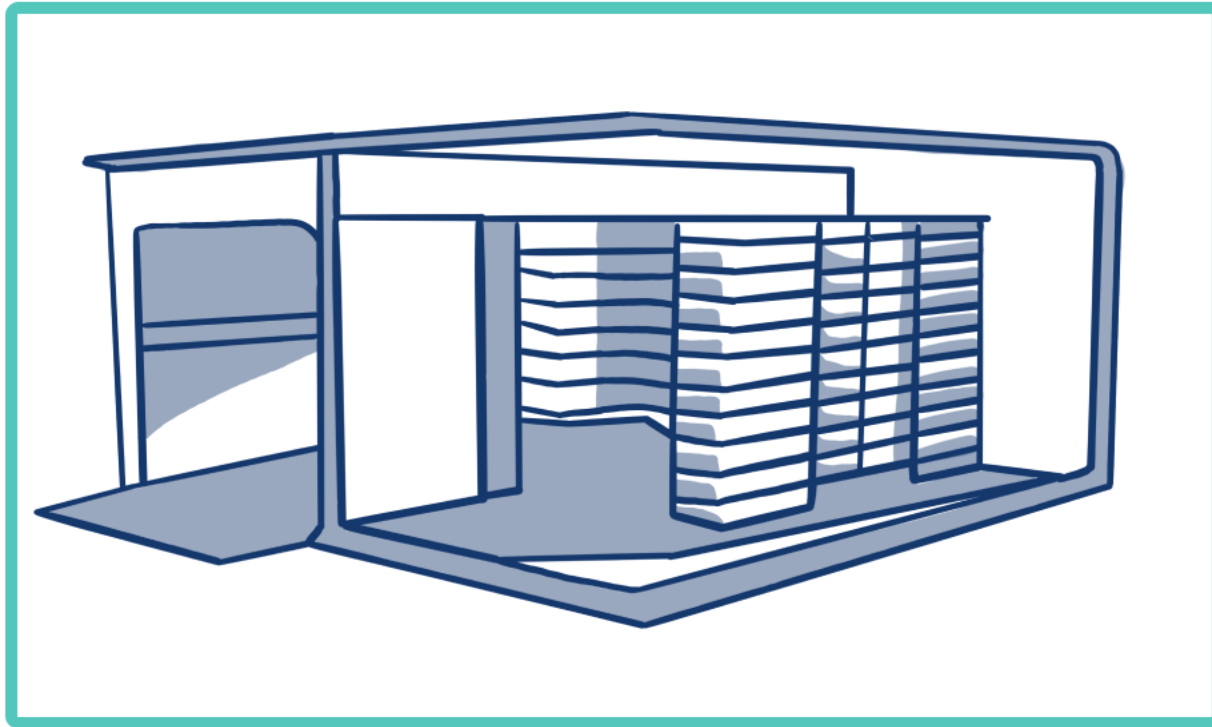
✈ UAM/AAM Landing Port

AAMs aligned with future goals:

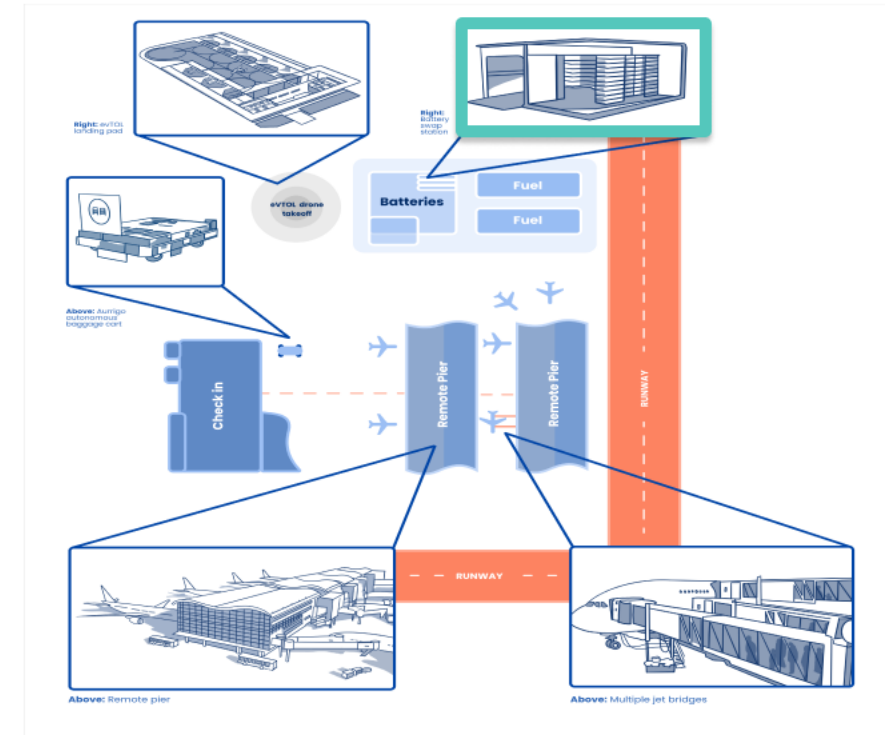
- Highly **automated** aircraft that operate at lower altitudes to carry passengers and cargo
- **Maximizing** logistic transportation and passenger movement
- **Increasing** electrification
- Integrating automation
- Further **research** on infrastructure, aircraft standards and public acceptance
- DEN has sufficient space to implement such a landing port



✈ Airport Innovations

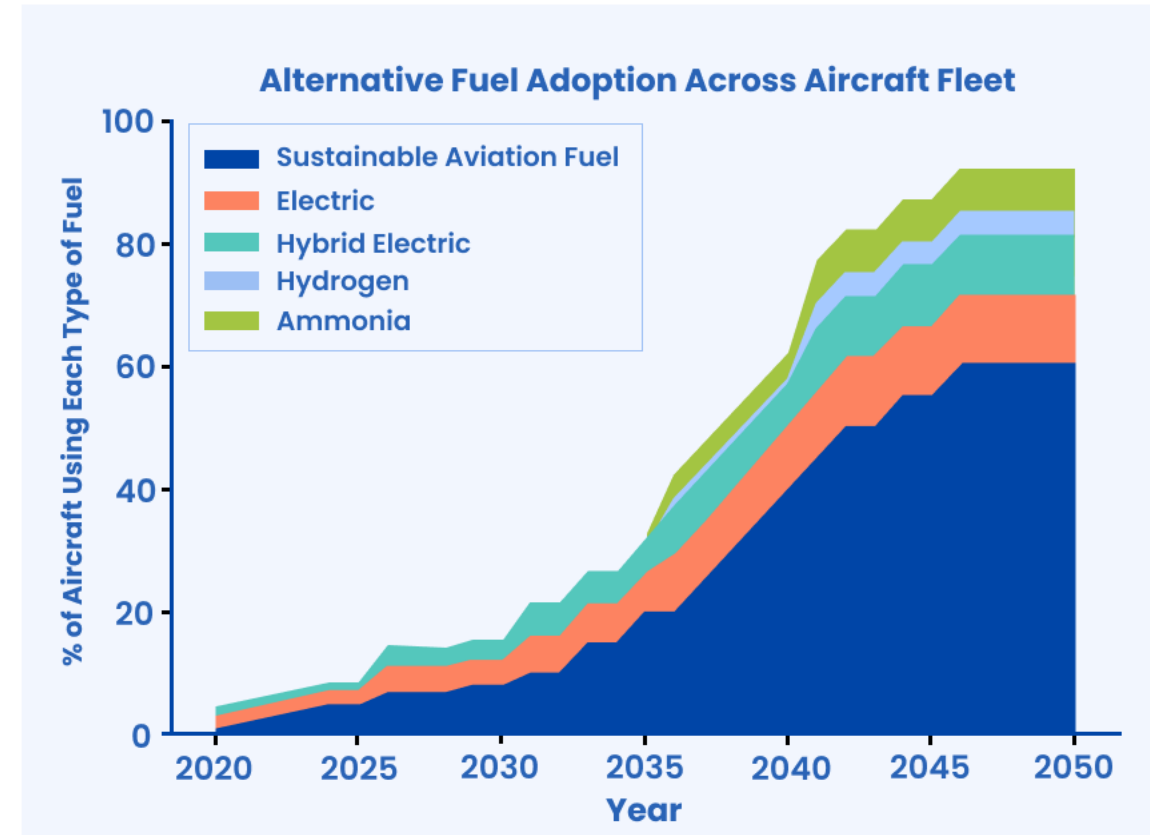


Centralized Energy Hub



✈️ 2050 Aviation Landscape: New Fuel Types

- We expect a **mixture** of new fuel sources
- Primarily composed of **Sustainable** Aviation Fuel
- We also expect an **expansion** of electric aircraft for short range flights
- May not reach 100% alternative fuels by 2050
- Can be **adapted** to reflect each airport's regional natural resources and public opinion drivers



✈ Centralized Energy Hub



**Sustainable
Aviation Fuels**

Jet fuel containers
repurposed to store
Sustainable Aircraft
Fuel mixtures

✈ Centralized Energy Hub



**Sustainable
Aviation Fuels**

**Hydrogen
Fuels**

Jet fuel containers
repurposed to store
Sustainable Aircraft
Fuel mixtures

Hydrogen containers
which account for **lower
volumetric density**
(pressurization, cooling)

✈ Centralized Energy Hub



**Sustainable
Aviation Fuels**

**Hydrogen
Fuels**

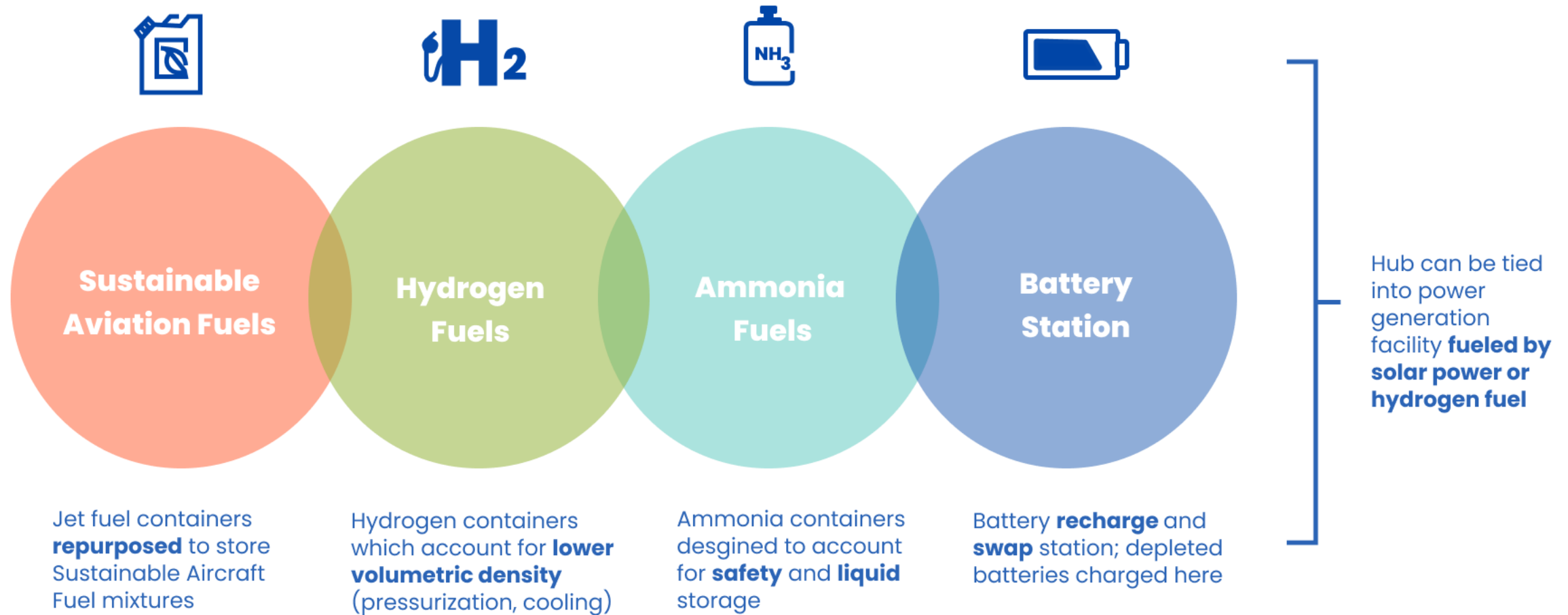
**Ammonia
Fuels**

Jet fuel containers
repurposed to store
Sustainable Aircraft
Fuel mixtures

Hydrogen containers
which account for **lower
volumetric density**
(pressurization, cooling)

Ammonia containers
designed to account
for **safety** and **liquid**
storage

✈ Centralized Energy Hub



KEY TAKEAWAYS

✈ Sustainability

Apply technologies both to make airport infrastructure adaptable and support new aircraft

- Remote Pier
- Gate Design
- UAM Landing Port
- Centralized Energy Hub

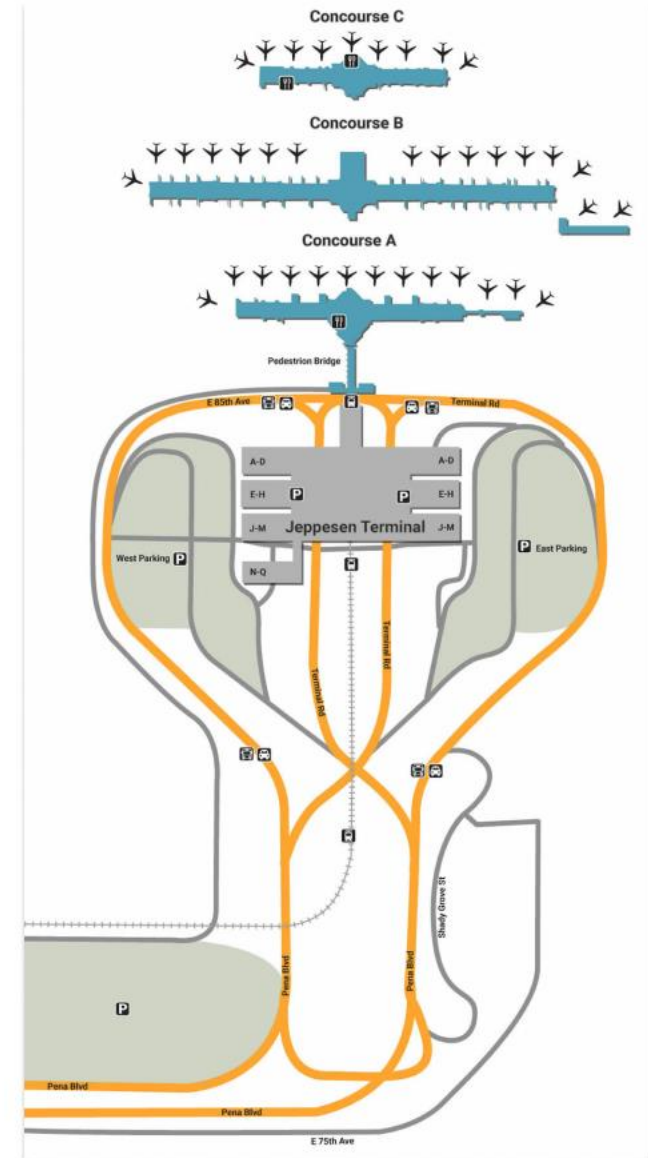
✈ Connected Autonomy

Incorporate autonomy into applicable airport tasks

- Integrated autonomous platform
- Standardized logistics containers

DENVER INTERNATIONAL AIRPORT

- DEN cites **sustainability** as one of its guiding principles
- A **hub** for both freight logistics and passengers (100 million annual passengers) -> logistic containers are great application
- New AEROS Logistics center - **connected autonomy** will help with operations



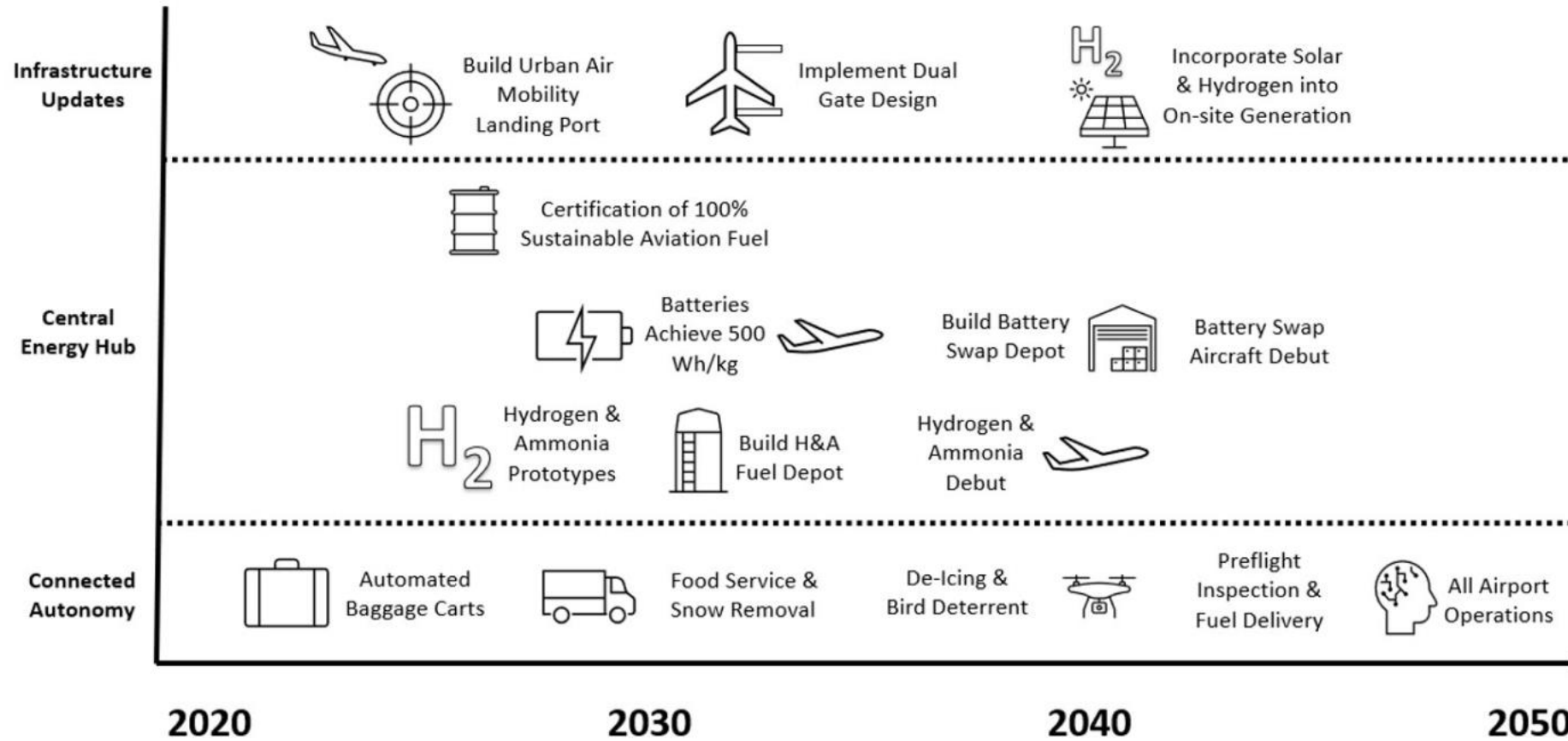
Left: Denver International Airport, n.d.
Right: Airport Shuttles, n.d.

DENVER INTERNATIONAL AIRPORT

- Energy hub can be **adapted** to Denver – make use of natural gas based hydrogen and available renewable energy
- Existing remote pier layout– **easy to expand on**
- Currently undergoing a gate expansion program
 - Gates will be long enough to fit **3 Airbus A380** aircraft



TECHNOLOGY DEVELOPMENT TIMELINE



CONCLUSIONS

- There are **many** sustainable aircraft solutions (fuel and form)
 - Select innovations that accomodate current and future customers and increase airport efficiency
- Assessed technology readiness **meets 2050 timeline**
 - Not all paths are the same, but can be properly sequenced
- Our Airport Innovations can be **applied to present** airports
 - As shown in our Denver case study

CONCLUSIONS

The US government and interested stakeholders should make a concerted effort on the **public policy front**

- **Develop** new technology standards
- **Inform** the public on the standards and safety of new aircraft
- **Establish** retraining and education for airport personnel and connected industry jobs
- **Support** technology and logistics chain development



Thank you!



Supporting Material

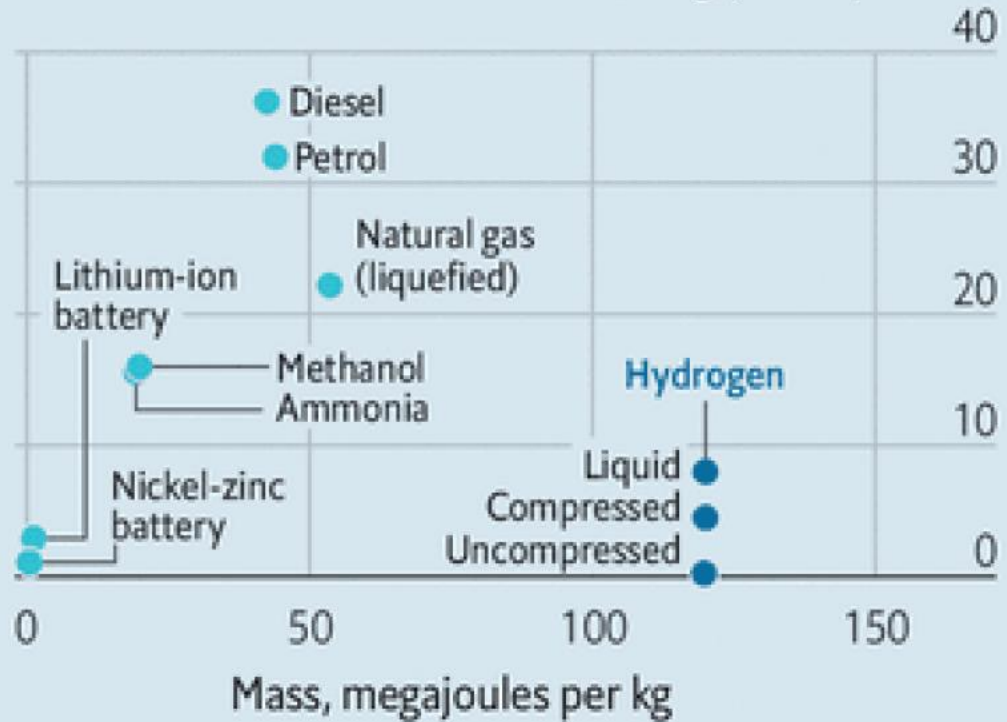
Energy Density of Fuels

Concentrated power

1

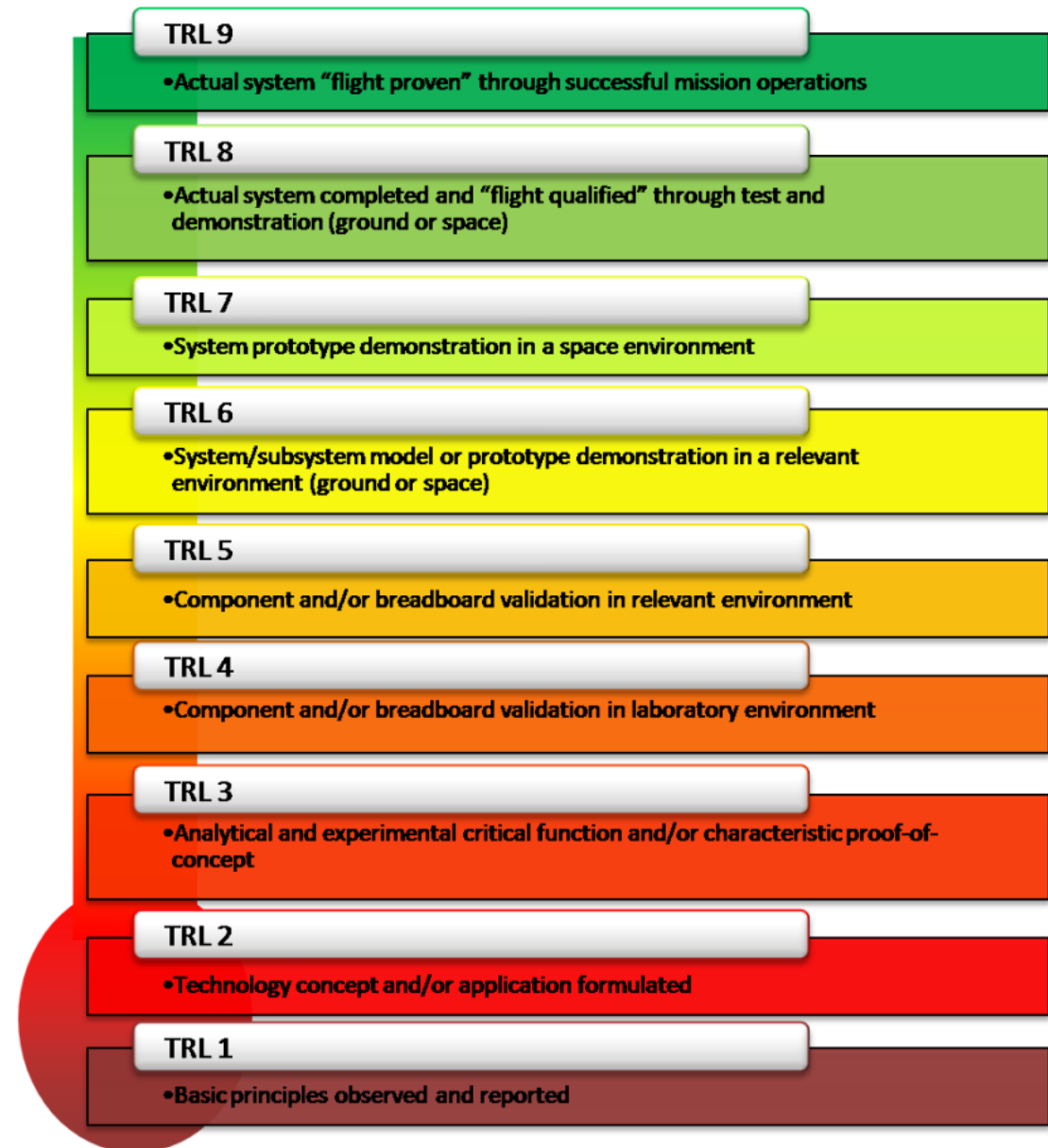
Energy density

Volume, megajoules per litre



Sources: DOE; Tran et al.; Vaclav Smil;
Schröder et al.; Valera-Medina et al.

Technology Readiness Levels



Sustainable Aviation Fuel

- + Carbon emission friendly, manufactured from recycled carbon materials
- + "Drop-in" fuel, does not require engine or storage modifications
- + Can be mixed with jet fuel
- + Tangible support from US Gov't, FAA, DOE, and commercial companies
- Limited feedstocks and manufacturing capability
- May never be cheaper than traditional jet fuel



Electric & Hybrid Electric

- + Zero emission energy source
- + Can combine with traditional combustion
- + Possibility of battery swap
- + Strong competitor for short range flights
- + Cool new air taxis!
- Low energy density relative to jet fuel
- Electricity infrastructure at remote airports



Hydrogen Fuel

- + Zero carbon emissions
- + Higher energy density than jet fuel
- Lower volumetric density than jet fuel
- Needs robust logistics chain
- Costly hydrogen generation process
- Public acceptance and FAA certification



Ammonia Fuel

- + Higher volumetric density and boiling point than hydrogen
- + Well established supply chain
- Produces Nitrous Oxides (NO_x)
- Early stage of technology development

